High-Resolution Remote Sensing Image Captioning Based on Structured Attention

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***Abstract*— Automatically generating language descriptions of remote sensing images has become an emerging research hot spot in the remote sensing field. Attention-based captioning, as a representative group of recent deep learning-based captioning methods, shares the advantage of generating the words while highlighting corresponding object locations in the image. Standard attention-based methods generate captions based on coarse-grained and unstructured attention units, which fails to exploit structured spatial relations of semantic contents in remote sensing images. Although the structure characteristic makes remote sensing images widely divergent to natural images and poses a greater challenge for the remote sensing image captioning task, the key of most remote sensing captioning methods is usually borrowed from the computer vision community without considering the domain knowledge behind. To overcome this problem, a fine-grained, structured attention-based method is proposed to utilize the structural characteristics of semantic contents in high-resolution remote sensing images. Our method learns better descriptions and can generate pixelwise segmentation masks of semantic contents. The segmentation can be jointly trained with the captioning in a unified framework without requiring any pixelwise annotations. Evaluations are conducted on three remote sensing image captioning benchmark data sets with detailed ablation studies and parameter analysis. Compared with the state-of-the-art methods, our method achieves higher captioning accuracy and can generate high-resolution and meaningful segmentation masks of semantic contents at the same time.**

***Index Terms*— Image captioning, image segmentation, remote sensing image, structured attention.**

I. Introduction

IMAGE captioning is an important computer vision task that emerged in recent years and aims to automatically generate language descriptions of an input image [1], [2]. In the remote sensing field, image captioning also has attracted increasing attention recently due to its broad application prospects both in civil and military usages, such as remote sensing image retrieval and military intelligence generation [3]. Different from other tasks in remote sensing field, such as object detection [4]–[8] classification [9]–[11], and segmentation [12]–[15], remote sensing image captioning focuses more on generating comprehensive sentence descriptions rather than predicting individual category tags or words. To generate accurate and detailed descriptions, the captioning model needs to not only determine the semantic contents that exist in the image but also have a good understanding of the relationship between them and what activities that they are engaged in [2].

In the most recent deep learning-based image captioning methods, the models are built based on the "encoder–decoder" network architecture [1], [2], [16]–[18]. In the encoding stage, deep convolutional neural networks (CNNs) are used to extract high-level internal representations of the input image. In the decoding stage, a recurrent neural network (RNN) is typically trained to decode the representations to sentence descriptions. More recently, the visual attention mechanism, a technique derived from automatic machine translation [19], [20], has greatly promoted the research progress in image captioning [21]–[26]. The attention mechanism was originally introduced to improve the performance of an RNN model by taking into account the input from several time steps to make one prediction [19]. In image captioning, visual attention can help the model better exploit spatial correlations of semantic contents in the image and highlight those contents while generating corresponding words [21].

For the remote sensing image captioning task, Qu *et al.* [27] first proposed a deep multimodal neural network model for high-resolution remote sensing image caption generation. Shi and Zou [3] proposed a fully convolutional networks (FCNs) captioning model, which mainly focuses on the multilevel semantics and semantic ambiguity problems. Lu *et al.* [28] explored several encoder–decoder-based methods and their attention-based variants and published a remote sensing image caption data set named RSICD. Wang *et al.* [29] introduced the multisentence captioning task and proposed a framework using semantic embedding to measure the image representation and the sentence representation to improve captioning results. Lu *et al.* [30] proposed a sound active attention (SAA) framework for more specific caption generation according to the interest of the observer. Wang *et al.* [31] proposed the retrieval topic recurrent memory network that first retrieves the topic words of input remote sensing images from the

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topic repository and then generates the captions by using a recurrent memory network [32] based on both the topic words and the image features. Ma *et al.* [33] proposed two multiscale captioning methods to grab multiscale information for generating better captions. Cui *et al.* [34] proposed an attention-based remote sensing image semantic segmentation and spatial relationship recognition method. However, the captioning module in their method just follows the classical model [21] without modification based on the characteristic of remote sensing images. The captioning module is independent of other modules, and the accuracy of caption generation is not improved by other modules. Sumbul *et al.* [35] proposed a summarization-driven image captioning method, which integrated the summarized ground-truth captions to generate more detailed captions for remote sensing images. Li *et al.* [36] proposed a truncation cross entropy to deal with the overfitting problem. Wang *et al.* [37] proposed a word–sentence framework to extract the valuable words first and then organize them into a well formed caption. Huang *et al.* [38] proposed a denoising-based multiscale feature fusion mechanism to enhance the image feature extraction. Li *et al.* [39] proposed a multilevel attention model to enhance the effect of attention through a hierarchical structure. Wu *et al.* [40] proposed a scene attention mechanism that tried to catch the scene information to improve the captions.

These remote sensing image captioning methods are all based on encoder–decoder architecture, which can be roughly divided into two groups: 1) methods without visual attention mechanisms that constructed between caption and image space [3], [27]–[29], [31], [35]–[38] and 2) methods with visual attention mechanisms [28], [30], [33], [34], [39], [40]. The visual attention mechanisms in these methods are designed based on coarse-grained, unstructured attention units, which fails to exploit structured spatial relations of semantic contents in remote sensing images. For example, in the popular natural image captioning method "Show, attend and tell" [21], the authors uniformly divide the image feature map into 14×14 spatial units. However, in remote sensing images, the semantic contents are usually highly structured where narrow and irregularly shaped objects, such as roads, rivers, and structures, usually occupy a large portion. The uniform division of the feature map inevitably leads to an underexploit of the spatial structure of remote sensing semantic contents. Besides, due to the coarse division of attention units, these methods also fail to produce fine-grained attention maps of irregularly shaped semantic contents.

In this article, we show that the structured and pixel-level regional information can be used to enhance the efficacy of attention-based remote sensing image captioning. In computer vision, in-depth research has been made on the pixel-level description of irregularly shaped semantic contents, where a representative group of the method is semantic segmentation [41]–[45]. We, thus, introduce a structured attention module in our captioning model and propose a joint captioning and segmentation framework for high-resolution remote sensing images by taking advantage of the structured attention mechanism. The structured attention module aims to focus on the semantic contents in the remote sensing images with structured geometry and appearance. Structured attention is performed on each structured unit obtained in the segmentation proposal generation, that is, the pixels within each structured unit receive the same attention weight, while different structured units get different attention weights. In this way, the proposed method can exploit the spatial structure of semantic contents and produce fine-grained attention maps to guide the decoder to more proper caption generation. We show our method generates better sentence descriptions and pixel-level object masks under a unified framework. It is worth mentioning that, in our method, the segmentation is trained solely based on the image-level ground-truth sentences and does not require any pixelwise annotations.

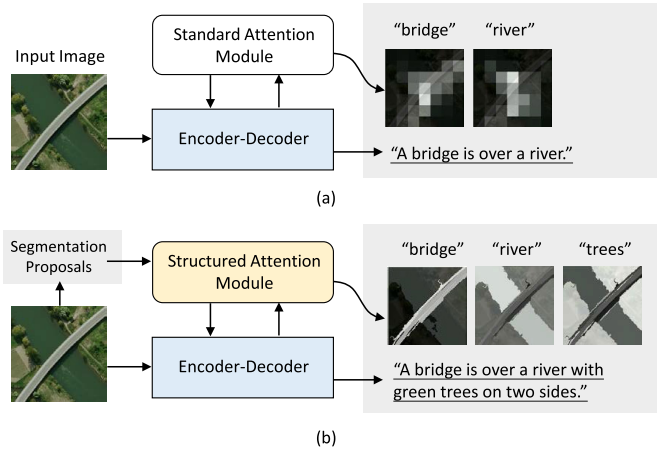


Fig. 1. Brief comparison between (a) standard attention-based captioning method and (b) proposed structured attention-based captioning method. In (a), the captions are learned from a set of coarse and unstructured image regions. As a comparison, in (b), our method exploits the fine-grained structure of the image and, thus, generates more accurate descriptions.

Fig. 1 shows the key differences between the proposed method and previous attention-based methods. In our method, we first divide the input image into a set of class-agnostic segmentation proposals and then encode the structure of each of the segmentation proposals into our attention module. Structured attention can guide the model to accurately focus on highly structured semantic contents during the training, thereby improving the performance of the image captioning task. Although the class label of each proposal is not available during the training and is considered as latent variables, we show the correspondence between the predicted words and the attention weights for each proposal can be adaptively learned under a weakly supervised training process. Our method, therefore, produces much more accurate attention maps for the semantic contents than those unstructured attention methods.

Extensive evaluations of our method are made on three benchmark data sets. Our method achieves higher captioning accuracy than other state-of-the-art captioning methods and generates object masks with high quality. Detailed ablation studies and parameter analysis are also conducted, which suggests the effectiveness of our method.

The contributions of this article are summarized as follows.

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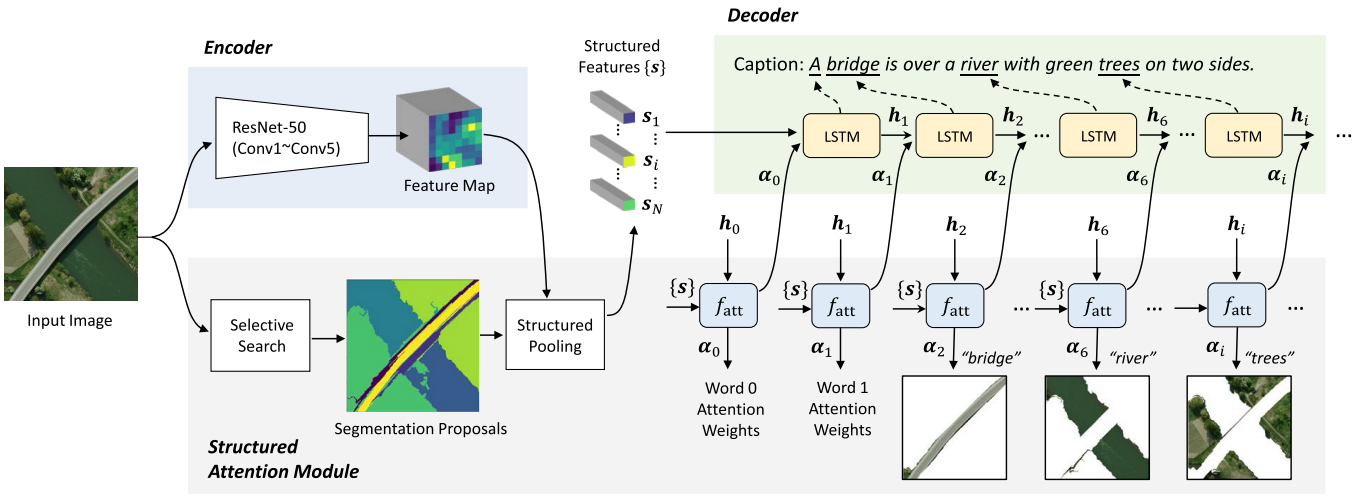


Fig. 2. Overview of the proposed captioning method. Our method consists of three parts: an encoder, which maps the input image to feature maps; a decoder, which generates the sentences based on the image feature; and a structured attention module, which interacts with the decoder during the captioning and at the same time generates pixel-level object masks.

1) We propose a novel image captioning method for high-resolution remote sensing images based on the structured attention mechanism. The proposed method deals with image captioning and pixel-level segmentation under a unified framework.
2) We investigate the possibility of using structured attention for weakly supervised image segmentation. To the best of our knowledge, such a topic has rarely been studied before.
3) We achieve higher captioning accuracy over other state-of-the-art methods on three remote sensing image captioning benchmark data sets.

The rest of this article is organized as follows. In Section II, we will introduce the structured attention and the details of our method. Experimental results and analysis are given in Section III. The conclusions are drawn in Section IV.

II. Methodology

In this section, we give a detailed introduction to the proposed structured attention method and how we build our image captioning model on top of it.

A. Overview of the Method

The captioning model proposed in this article mainly consists of three parts: an encoder, a decoder, and a structured attention module. Fig. 2 shows the processing flow of the proposed model. From the natural image captioning literature, we borrow the encoder–decoder framework that has been shown to work well in the image captioning task. We use a deep CNN as our encoder to extract high-level feature representations from the input image. It is worth mentioning that our method is, indeed, independent of the choice of the backbone model. Any deep CNN can be used as an encoder. We use a long short-term memory network [46], [47] as our decoder to decode the image features to the sentence description. Before feeding features to the structured attention module, we use a predefined method "Selective Search" [48] to segment the input image to a set of class-agnostic segmentation proposals based on the color and texture features. The selective search module in our framework requires the remote sensing image to be high resolution to extract available segmentation proposals. Fig. 3 shows some samples generated by the selective search. The proposals are then synchronously encoded to our attention module with the image features by using a newly proposed pooling method, named the "structured pooling" method. In this way, the original image features are recalibrated, and we, thus, can obtain a set of structured region descriptions for captioning and mask generation. The attention weights generated by the model for each region on predicting a certain word are considered as the probability that the region belongs to the word category (e.g., building, tree, and bridge).

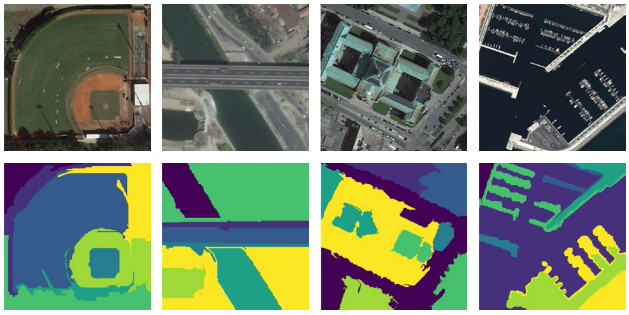


Fig. 3. Input images (first row) and the class-agnostic segmentation proposals generated by the selective search method (second row).

B. Encoder and Decoder

We use the 50-layer deep residual network (ResNet-50) [49] as our encoder. We remove the full connection layer (prediction layer) of the ResNet-50 and use the feature maps produced by the last convolution block "Conv_5" as our internal feature representations. Our decoder is a one-layer LSTM with

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512 hidden units. The LSTMs are a special kind of RNN, capable of learning long-term dependences. By selectively forgetting and updating information in the training process, the LSTM can achieve better performance in the complex sequential prediction problems than the vanilla RNNs. Our decoder is trained to generate the word score vector $\mathbf{y}_t \in \mathbb{R}^K$ at each time step t based on a context vector $\mathbf{z}_t$, a previous hidden state vector $\mathbf{h}_{t-1}$ and a previously generated word score vector $\mathbf{y}_{t-1}$, where K is the size of the vocabulary. The prediction of $\mathbf{y}_t$ can be written as follows:

$$\mathbf{y}_t = \mathbf{L}_o\left(\mathbf{L}_h\mathbf{h}_t + \mathbf{L}_y\mathbf{y}_{t-1} + \mathbf{L}_z\mathbf{z}_{t-1}\right) \tag{1}$$

where $\mathbf{L}_h$, $\mathbf{L}_y$, and $\mathbf{L}_z$ are a group of trainable parameters that transform the input vectors to calibrate their dimensions. The $\mathbf{L}_o$ is a group of trainable parameters that transform from the summarized vectors to the output score vectors. To compute $\mathbf{h}_t$, $\mathbf{z}_t$, and $\mathbf{y}_t$ at each time step, the LSTM gates and internal states are defined as follows:

$$\begin{aligned}
\mathbf{i}_t &= \sigma\left(\mathbf{W}_i\mathbf{x}_t + \mathbf{b}_i\right) \\
\mathbf{f}_t &= \sigma\left(\mathbf{W}_f\mathbf{x}_t + \mathbf{b}_f\right) \\
\mathbf{o}_t &= \sigma\left(\mathbf{W}_o\mathbf{x}_t + \mathbf{b}_o\right) \\
\tilde{\mathbf{c}}_t &= \tanh\left(\mathbf{W}_c\mathbf{x}_t + \mathbf{b}_c\right) \\
\mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \\
\mathbf{h}_t &= \mathbf{o}_t \odot \tanh\left(\mathbf{c}_t\right)
\end{aligned} \tag{2}$$

where $\mathbf{x}_t$ is the concatenation of previous hidden state $\mathbf{h}_{t-1}$, and the previously generated word vector $\mathbf{y}_{t-1}$ and the context vector $\hat{\mathbf{z}}_t$ are $\mathbf{x}_t = [\mathbf{h}_{t-1}; \mathbf{P}\mathbf{y}_{t-1}; \hat{\mathbf{z}}_t]$. $\mathbf{P} \in \mathbb{R}^{m\times K}$ is an embedding matrix, where m denotes the embedding dimension. $\mathbf{i}_t$, $\mathbf{f}_t$, $\mathbf{o}_t$, $\mathbf{c}_t$, and $\mathbf{h}_t$ are the outputs of the input gate, forget gate, output gate, memory, and hidden state of the LSTM, respectively. $\mathbf{W}_i$, $\mathbf{W}_f$, $\mathbf{W}_o$, and $\mathbf{W}_c$ are trainable weight matrices, and $\mathbf{b}_i$, $\mathbf{b}_f$, $\mathbf{b}_o$, and $\mathbf{b}_c$ are their trainable biases. $\sigma(\cdot)$, $\tanh(\cdot)$ and $\odot$ represent the logistic sigmoid activation, hyperbolic tangent function and elementwise multiplication operation, respectively.

Finally, to produce word probabilities $\mathbf{p}_t$, we use a "softmax" layer to normalize the generated score vectors to probabilities

$$\begin{aligned}
\mathbf{p}_t &= \text{softmax}(\mathbf{y}_t) \\
&= \exp(\mathbf{y}_t) \Big/ \sum_{i=1}^{K} \exp\left(y_t^{(i)}\right).
\end{aligned} \tag{3}$$

C. Structured Attention

1) Structured Pooling: Most CNNs produce unstructured image feature representations. Here, we propose a new pooling operation called "structured pooling" to generate structured feature representations given a set of region proposals of any shapes. The structured pooling can be considered as a modification of the standard "region of interest (ROI) pooling." The difference between the two operations is that the ROI pooling is only designed to pool the features from rectangular regions, while the structured pooling applies to the regions of any shape. Suppose that I is the input image and $\mathbf{F} \in \mathbb{R}^{h\times w\times c}$ represent the image features produced by the encoder, where h, w, and c are the height, width, and the number of channels, respectively. The region proposals R_i, $i = 1, \ldots, N$ produced by the selective search are considered as the base units when performing structured pooling.

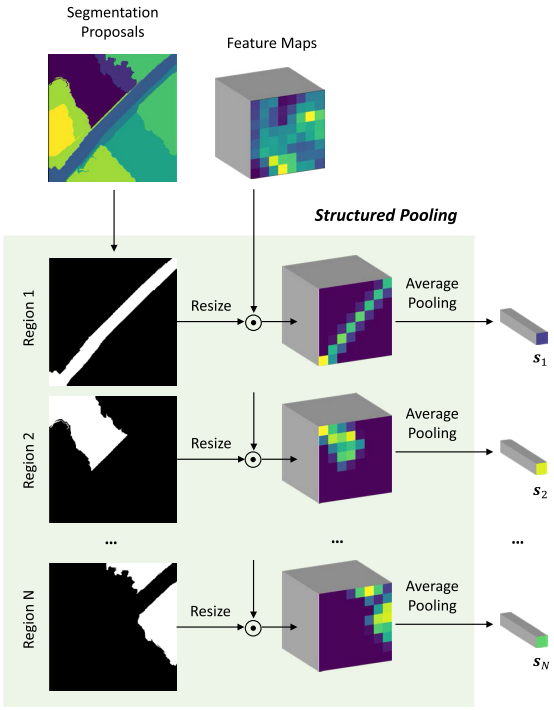


Fig. 4. Illustration of the proposed structured pooling method. The notation $\odot$ represents the elementwise product operation.

For the unit i, the structured feature representation $\mathbf{s}_i$ produced by the structured pooling can be represented as follows:

$$\mathbf{s}_i = \frac{1}{hw} \sum_{(x,y)\in R_i'} \mathbf{F}(x, y) \odot R_i' \tag{4}$$

where R_i' is the projected region proposal, which is resized from the size of the input image to the size of the feature map. The summation is performed among the pixels (x, y) within the region R_i' along the spatial dimensions. It should be noticed that, when we average the feature values within a certain region R_i', we divide the number of all spatial pixels in the feature map (hw) instead of the number of valid pixels in that region. The reason behind this is that we want to enhance the features according to their structured unit size, that is, the features of small structured units will be less weighted to reduce the noise effect from these regions.

Fig. 4 gives a simple illustration of the proposed structured pooling operation. To help understand, in this figure, we show an alternative but equivalent way of performing structured pooling, where we first pixelwisely multiply the features on a set of resized region masks and then perform the global average pooling to produce the pooling output. To reduce the misalignment effect, we use the bilinear interpolation when we reduce the size of the binary region masks.

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It is worth noting that, although proposal-based approaches are widely used in computer vision tasks, the proposed structured attention method is designed specifically for remote sensing images. Since remote sensing images are captured from high above, many semantic contents in remote sensing images, such as rivers and bridges, show highly structured characteristics. For example, bridges are always long and straight, and rivers are always winding and slender, while buildings are mostly in the form of regular polygon aggregation. As a comparison, semantic contents in natural images often lack regular and structured shape outlines under different views and occlusions. Since the proposed structured attention mechanism relies on effective structure extraction of the semantic contents in images, it is more suitable for remote sensing images rather than natural images.

2) Context Vector Generation: The context vector $\hat{\mathbf{z}}_t$ in our method is a dynamic representation of the corresponding structured unit of the image at the time t. Given the feature representations $\mathbf{s}_i$ and the previous hidden state $\mathbf{h}_{t-1}$, we calculate an attention weight value $\alpha_t^{(i)}$, which represents the degree of correlation between the structured units and the generated word vector $\mathbf{y}_t$

$$\tilde{\alpha}_t^{(i)} = f_{\text{att}}(\mathbf{s}_i, \mathbf{h}_{t-1}) \tag{5}$$

where $f_{\text{att}}(\cdot)$ represents a multilayer perceptron (MLP), which is trained to generate the attention weights.

To build the network $f_{\text{att}}(\cdot)$, we first adjust the dimensions of $\mathbf{s}_i$ and $\mathbf{h}_{t-1}$ to the same number by passing each of them through a fully connected layer. Then, the transformed vectors are added together to fuse the information from both the structured unit and the context, and the fusion vector is further fed to another fully connected layer to produce the attention weight $\alpha_t^{(i)}$

$$f_{\text{att}}(\mathbf{s}_i, \mathbf{h}_{t-1}) = f_3(\text{ReLU}(f_1(\mathbf{s}_i) + f_2(\mathbf{h}_{t-1}))) \tag{6}$$

where f_1, f_2, and f_3 represent the three fully connected layers and ReLU$(\cdot)$ represents the rectified linear unit activation function. Then, the attention weights of the N unique regions at the time step t are normalized with a softmax layer to produce the final attention vector $\boldsymbol{\alpha}_t$

$$\boldsymbol{\alpha}_t = \text{softmax}\left(\left[\tilde{\alpha}_t^{(1)}, \ldots, \tilde{\alpha}_t^{(N)}\right]\right). \tag{7}$$

Once we get the attention weighted vector, the context vector $\mathbf{z}_t$ can be finally computed as a linear combination of the structured feature represents $\mathbf{s}_i$ and their attention weights $\alpha_t^{(i)}$

$$\mathbf{z}_t = \sum_{i=1}^{N} \alpha_t^{(i)} \mathbf{s}_i. \tag{8}$$

Note that, at the time step t, the attention weight of each structured unit is computed based on the same context information $\mathbf{h}_{t-1}$, which ensures that the initial competitiveness of each structured unit is fair and reduces the possibility of introducing deviation. This is called the "context information broadcast" mechanism, which was introduced by Vinyals *et al.* [1] for the first time.

3) Object Masks' Generation: We generate the object masks based on the attention weights of each structured region. In our attention module, the attention weights $\alpha_t^{(i)}, i = 1, \ldots N$ represent the semantic correlation between the tth word of the sentence and each of the N regions. The larger the $\alpha_t^{(i)}$, the more relevant it is to the ith structured unit R_i. We use the attention weights as the category probability of the segmentation output. The nouns of the semantic contents of interest can be easily picked out from the generated captions by comparing each word to a predefined noun set. The pixelwise object masks can be finally generated by binarizing the segmentation weights of each region.

D. Loss Functions

Since image captioning is a sequential prediction problem of each word in the sentence, we follow the previous works [1], [21] and formulate the prediction of each word as a regularized classification process. The loss can, thus, be written as a running sum of the regularized cross-entropy loss of each word in the sentence

$$\mathcal{L}(\mathbf{x}) = -\sum_{t=1}^{C} \log\left(\sum_{j=1}^{K} \hat{y}_t^{(j)} p_t^{(j)}\right) + \beta r_d(\boldsymbol{\alpha}_t) + \gamma\, r_v(\boldsymbol{\alpha}_t) \tag{9}$$

where $\mathbf{p}_t = [p_t^{(1)}, \ldots, p_t^{(K)}]$ is the predicted word probability vector. $\hat{\mathbf{y}}_t = [\hat{y}_t^{(1)}, \ldots, \hat{y}_t^{(K)}]$ is the one-hot label of the tth word in the ground-truth caption, and $\hat{y}_t^{(j)} \in \{0, 1\}$. C is the number of words in the generated sentence. $r_d(\boldsymbol{\alpha}_t)$ and $r_v(\boldsymbol{\alpha}_t)$ are the doubly stochastic regularization (DSR) [21] and the proposed attention variance regularization (AVR), which we will introduce later. β and γ are the weight coefficients for balancing different loss terms.

1) Doubly Stochastic Regularization: In Section II-C2, we show that $\sum_i \alpha_{t,i} = 1$ since the attention weights are finally normalized by a softmax function. Here, we further regularize the attention weights from the time dimension and introduce the DSR as follows:

$$r_d(\boldsymbol{\alpha}_t) = \sum_{i=1}^{N}\left(1 - \sum_{t=1}^{C} \alpha_{t,i}\right)^2. \tag{10}$$

This regularization term encourages the model to pay equal attention to each part of the image during the generation of captions. In other words, it can prevent some regions from always receiving strong attention, while it can prevent other regions from being ignored all the time.

2) Attention Variance Regularization: When we fixed the time step t and look at the attention weights of each structure regions, we usually hope to see these regions receive a highly diverse attentions. This means that we do not want each region receive equal attentions. We, thus, design the AVR term to enforce the regions have a high variance in their attention weights

$$\begin{aligned} r_v(\boldsymbol{\alpha}_t) &= -\sum_{t=1}^{C} \|\boldsymbol{\alpha}_t - \mathbb{E}\{\boldsymbol{\alpha}_t\}\|_2^2 \\ &= -\sum_{t=1}^{C} \left\|\boldsymbol{\alpha}_t - \frac{1}{N}\right\|_2^2 \end{aligned} \tag{11}$$

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where we have $\mathbb{E}\{\boldsymbol{\alpha}_t\} = 1/N$ since the $\boldsymbol{\alpha}_t$ has been normalized by the softmax function. We take the negative value of the l_2 norm since we want to maximize the attention variance. It is easy to proof that, when the $\boldsymbol{\alpha}_t$ is an one-hot vector, i.e., only one region contributes to the prediction of the current word, the value of $r_v(\boldsymbol{\alpha}_t)$ will reach its minimum. On the contrary, when every region receives equal attention, i.e., $\alpha_t^{(i)} = 1/N$, $i = 1, \ldots, N$, in which we do not hope to see, $r_v(\boldsymbol{\alpha}_t)$ will be maximized.

E. Implementation Details

1) Training Details: In the training phase, we use the Adam optimizer [50] to train our model. We set regularization coefficients $\beta = \gamma = 1$. We set the learning rate of our encoder to $1e^{-4}$ and set the learning rate of our decoder learning rate to $4e^{-4}$. The batch size is set to 64, and the model is trained for 100 epochs. In our encoder, the ResNet-50 is pretrained on the ImageNet data set [51]. To speed up training, we only fine-tune the convolutional blocks 2–4 of the ResNet-50 during training. In our decoder, the memory and hidden state gate of the LSTM at the time step 0 are initialized separately based on the averaged image features. We use a fully connected layer to transform the features to produce their 0-time inputs.

2) Segmentation Proposal Generation: When we use the selective search to generate segmentation proposals, three key parameters need to be specifically tuned, including a smooth parameter σ of the Gaussian filter, a min_size parameter, which controls the minimum bounding box size of the proposals, and a scale parameter s, which controls the initial segmentation scales. We set $\sigma = 0.8$, min_size $= 100$, and $s = 100$. Besides, to prevent oversegmentation, we applied the guided image filter [52] to preprocess the image before the selective search. The guided image filter can effectively smooth the input image while keeping its edge and structures. The smoothed images are only used for generating segmentation proposals. When the encoder computes the image features, we still use the original images.

3) Beam Search: At the inference stage, instead of using a greedy search that chooses the word with the highest score and uses it to predict the next word, we apply the beam search [53] to generate more stabilized captions. The beam search selects the top-k candidates in each time step and then predicts top-k new words accordingly for each of these sequences in the next step. Then, the new top-k sequences of the next time step are selected out of all $k \times k$ candidates. It is worth mentioning that, in consideration of computational efficiency, top-k candidate sequences are selected for each time step, and the sequence with the highest score is selected as the final caption output at the last time step. Therefore, up to time t, k sequences are generated instead of k^t. k is called the "beam size," which is set to 5 for our experiment.

III. Experiments

In this section, we will introduce in detail our experimental data sets, metrics, and comparison results. We also provide ablation experiments, parameter analysis, and speed analysis to verify the effectiveness of the proposed structured attention module.

A. Experimental Setup

1) Data Sets: We conduct experiments on three widely-used remote sensing image captioning data sets: UCM-Captions [27], Sydney-Captions [27], and RSICD [28]. For each data set, we followed the standard protocols on splitting the data set into training, validation, and test sets. In any of the three data sets, each image is labeled with five sentences as ground-truth captions. The following are the details of the three data sets.

a) UCM-Captions: The UCM-Captions data set [27] is built based on the UC Merced land use data set [54]. It contains 2100 remote sensing images from 21 types of scenes. Each image has a size of 256 × 256 pixels and a spatial resolution of 0.3 m/pixel.

b) Sydney-Captions: The Sydney-Captions data set [27] is built based on the Sydney land data set [55]. It totally contains 613 remote sensing images collected from the Google Earth imagery in Sydney, Australia. Each image has a size of 500 × 500 pixels and a spatial resolution of 0.5 m/pixel.

c) RSICD: The RSICD [28] is the most widely used data set for remote sensing image caption generation task. It contains 10 921 remote sensing images collected from the AID data set [56] and other platforms, such as Baidu Map, MapABC, and Tianditu. The images are in various spatial resolutions. The size of each image is 224 × 224 pixels.

2) Evaluation Metrics: We use four different metrics to evaluate the accuracy of the generated captions, including the bilingual evaluation understudy (BLEU) [57], ROUGE-L [58], METEOR [59], and CIDEr-D [60], which are all widely used in recent image captioning literature.

a) BLEU: The BLEU [57] measures the co-occurrences between the generated caption and the ground truth by using n-grams (a set of n ordered words). The key of the BLEU-n ($n = \{1, 2, 3, 4\}$) is the n-gram precision—the proportion of the matched n-grams out of the total number of n-grams in the evaluated caption.

b) METEOR: Since the BLEU does not take the recall into account directly, to address this weakness, the METEOR [59] is introduced to compute the accuracy based on explicit word-to-word matches between the captioning and the ground truth.

c) ROUGE-L: ROUGE-L [58] is a modified version of ROUGE, which computes an F-measure with a recall bias using the longest common subsequence (LCS) between the generated and the ground-truth captions.

d) CIDEr-D: CIDEr-D [60] is an improved version of CIDEr, which first converts the caption into the form of the term frequency inverse document frequency (TF-IDF) vector [61] and then calculates the cosine similarity of the reference caption and the caption generated by the model. CIDEr-D penalizes the repetition of specific n-grams beyond the number of times they occur in the reference sentence.

For any of the above four metrics, a higher score indicates a higher accuracy. The scores of BLEU, ROUGE-L, and METEOR are between 0 and 1.0. The score of CIDEr-D is between 0 and 10.0.

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TABLE I

ABLATION STUDIES ON THE PROPOSED STRUCTURED ATTENTION MECHANISM. THE EVALUATION SCORES (%) ARE REPORTED ON THE UCM-CAPTIONS DATA SET [27]

Soft Attention [21]	Structured Attention (ours)	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
✓	×	83.21	76.78	71.09	66.02	42.93	77.63	314.78
×	✓	**85.38**	**80.35**	**75.72**	**71.49**	**46.32**	**81.41**	**334.89**

TABLE II

ABLATION STUDIES ON THE TWO REGULARIZATION TERMS: DSR AND AVR. THE EVALUATION SCORES (%) ARE REPORTED ON THE UCM-CAPTIONS DATA SET [27]

DSR	AVR	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
×	×	80.45	74.48	69.64	64.99	42.68	76.89	302.06
✓	×	84.71	79.00	74.24	69.71	45.27	79.97	329.62
×	✓	83.73	77.64	72.74	68.21	44.02	78.79	312.28
✓	✓	**85.38**	**80.35**	**75.72**	**71.49**	**46.32**	**81.41**	**334.89**

TABLE III

EVALUATION SCORES (%) OF OUR METHODS WITH A DIFFERENT NUMBER OF PROPOSALS PER IMAGE. ALL MODELS ARE TRAINED AND EVALUATED ON THE UCM-CAPTIONS DATA SET [27]

Region num	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Training Time
4	84.63	78.84	73.85	69.40	45.62	80.59	328.23	155min
8	85.38	**80.35**	**75.72**	**71.49**	46.32	81.41	334.89	175min
12	85.70	80.03	75.27	70.75	**46.68**	**81.99**	332.93	193min
16	**85.78**	79.95	74.96	70.35	46.37	81.38	**338.80**	210min

B. Ablation Studies

The ablation studies are conducted to analyze the importance of three different technical components of the proposed method, including the structured attention module, DSR, and AVR. The ablation studies and parameter analysis experiments are performed on the UCM-Captions data set, the Sydney-Captions data set, and the RSICD data set. We found that the proposed method behaves similarly on these data sets. For brevity, we only report results on the UCM-Captions.

We first remove the proposed structured attention module of our method and replace it with a standard soft-attention module [21] while keeping other configurations unchanged. Table I shows the comparison results. The best scores are marked as bold. The results show that, compared with the baseline method, the structured attention improves the accuracy with a large margin in terms of all evaluation metrics (+5.47% on BLEU-4, +3.39% on METEOR, +3.78% on ROUGE-L, and +20.11% on CIDEr-D).

We then gradually remove the regularization terms from our loss function and train the corresponding captioning model separately. Table II shows their evaluation accuracy. We show that the DSR and the AVR can both yield noticeable improvements in captioning accuracy. Particularly, the proposed method trained with both of these two regularization terms achieves the best accuracy on all metrics.

C. Parameter Analysis

We also analyze two important parameters in our method: 1) the number of segmentation proposals N and 2) the beam size k.

We set the number of segmentation proposals N in the selective search to 4, 8, 12, and 16, train each model, and then evaluate the captioning accuracy accordingly. Table III shows the accuracy of our model with different segmentation proposals. The result shows that, when the number of regions increases from 4 to 8, the evaluation scores are greatly improved. However, when the number of regions further increases from 8 to 16, the improvement of evaluation scores becomes less significant, and some scores even decrease. This is because, when the number of segmentation proposals set by N is much larger than the actual number of regions in the remote sensing image, the oversegmentation of the image will destroy the structures of the semantic contents. We also report the models' training time in the last column of Table III. We show that increasing the number of regions leads to an increase in training time. To balance the accuracy of different metrics, we finally set the number of regions to $N = 8$.

The beam size k will affect the captioning accuracy and the inference time. We set different beam sizes in our method and analyze their accuracy and speed. All models are trained and evaluated on the UCM-Captions data set [27]. Table IV shows the evaluation results of our method with different beam sizes. We can see that, when the beam size increases from 1 to 5, the evaluation scores are improved but are saturated at 6. We also show that increasing the beam size leads to a slower inference speed. To balance the captioning accuracy and the inference speed, we set the beam size to $k = 5$.

Figs. 5 and 6 show the percentage of accuracy improvement of the proposed method with different numbers of segmentation proposals and different beam sizes. The percentage

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TABLE IV
EVALUATION SCORES (%) OF OUR METHODS WITH DIFFERENT BEAM SIZES. ALL MODELS ARE EVALUATED ON THE UCM-CAPTIONS DATA SET [27]

Beam Size	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	Inference speed
1	83.48	76.60	70.78	65.51	43.15	78.93	321.25	3.23
2	85.27	79.73	74.70	70.11	45.72	81.01	332.15	2.08
3	85.21	79.95	75.14	70.71	45.89	81.15	333.49	1.59
4	85.33	80.26	75.64	71.42	46.25	81.36	334.30	1.20
5	**85.38**	**80.35**	75.72	71.49	46.32	**81.41**	**334.89**	1.09
6	85.28	80.34	**75.73**	**71.54**	**46.33**	81.23	333.93	0.94

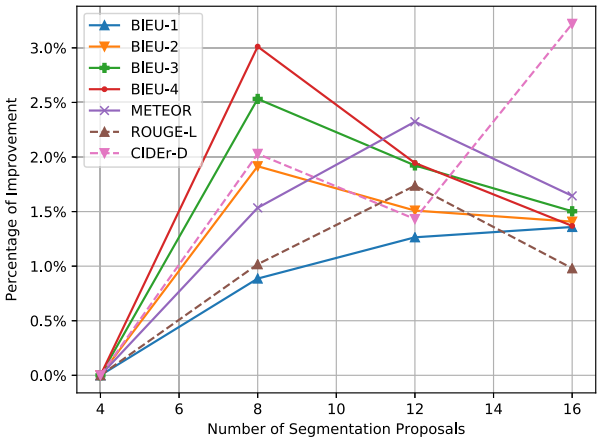


Fig. 5. (Better viewed in color) Percentage of accuracy improvement of the proposed method with a different number of segmentation proposals $N = \{4, 6, 8, 10, 12, 16\}$. All models are trained and evaluated on the UCM-Captions data set [27].

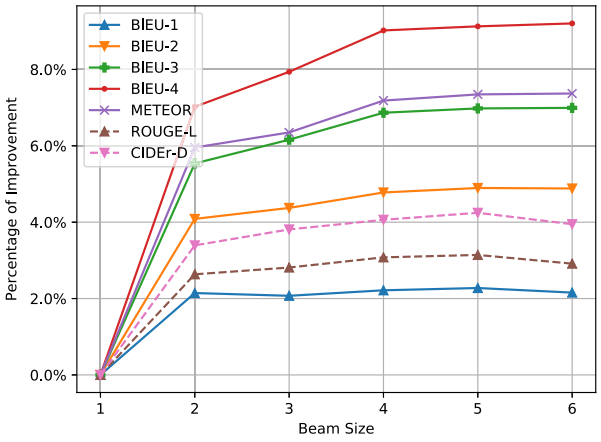


Fig. 6. (Better viewed in color) The percentage of accuracy improvement of the proposed method with different beam sizes $k = \{1, 2, 3, 4, 5, 6\}$. All models are evaluated on the UCM-Captions data set [27].

of improvement is defined as follows: $(\text{acc} - \text{acc}_0)/(\text{acc}_0) \times 100\%$. We define the accuracy on $N = 4$ and $k = 1$ as the baseline accuracy acc_0.

D. Comparison With Other Methods

In this section, we evaluate our method on three data sets and compared our method with a variety of recent image captioning methods. The comparison methods include the VLAD + RNN [28], VLAD + LSTM [28], mRNN [27], mLSTM [27], mGRU [62], mGRU-embedword [30], ConvCap [63], Soft-attention [28], Hard-attention [28], CSMLF [29], RTRMN [31], and SAA [30]. Among these methods, most of them (except mGRU and ConvCap) are initially designed for the remote sensing image captioning task. However, their basic ideas are mainly borrowed from the natural image captioning [1], [21]. The details of these models are described as follows.

1) VLAD + RNN: VLAD + RNN [28] uses the handcrafted feature descriptor "VLAD" [64] as its encoder to compute image representations and use a naive RNN as its decoder to generate captions.

2) VLAD + LSTM: VLAD + LSTM [28] also uses VLAD to compute the image features, but the difference is that it uses an LSTM as its decoder.

3) mRNN, mLSTM, and mGRU: These three methods [27], [27], [62] all use the VGG-16 [65] as their encoders but use different RNNs (naive RNN, LSTM, and GRU) as their decoders.

4) mGRU-Embedword: Similar to the mGRU [62], the mGRU-embedword [30] also uses the VGG-16 as its encoder and the GRU as its decoder. The difference is that mGRU-embedword uses a pretrained global vector, namely, GloVe [66], to embed words.

5) ConvCap: The ConvCap [63] uses the VGG-16 as its encoder and computes the attention weights based on the activations of the last convolutional layer. Instead of using the RNN-based decoder, this method generates captions by using a CNN-based decoder [63].

6) Soft-Attention and Hard-Attention: Soft-attention [28] and Hard-attention [28] are two methods using VGG-16 as the encoder and LSTM as the decoder. The decoders are build based on soft attention and hard attention mechanism [21], respectively.

7) CSMLF: CSMLF [29] is a retrieval-based method that uses latent semantic embedding to measure the similarity between the image representation and the sentence representation in a common semantic space.

8) RTRMN: RTRMN [31] uses Resnet-101 as its encoder and then uses the topic extractor to extract topic information. A retrieval topic recurrent memory network is used to generate captions based on the topic words. "RTRMN (semantic)" and "RTRMN (statistical)" are two variants of the RTRMN, which are based on semantic topics repository and statistical topics repository, respectively.

9) SAA: SAA [30] introduces an SAA framework to combine the sound information during the generation of captions.

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TABLE V
EVALUATION SCORES (%) OF DIFFERENT METHODS ON THE UCM-CAPTIONS DATA SET [27]

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
VLAD+RNN [28]	63.11	51.93	46.06	42.09	29.71	58.78	200.66
VLAD+LSTM [28]	70.16	60.85	54.96	50.30	34.64	65.20	231.31
mRNN [27]	60.10	50.70	32.80	20.80	19.30	-	214.00
mLSTM [27]	63.50	53.20	37.50	21.30	20.30	-	222.50
mGRU [62]	42.56	29.99	22.91	17.98	19.41	37.97	124.82
mGRU-embedword [30]	75.74	69.83	64.51	59.98	36.85	66.74	279.24
ConvCap [63]	70.34	56.47	46.24	38.57	28.31	59.62	190.15
Soft-attention [28]	74.54	65.45	58.55	52.50	38.86	72.37	261.24
Hard-attention [28]	81.57	73.12	67.02	61.82	42.63	76.98	299.47
CSMLF [29]	36.71	14.85	7.63	5.05	9.44	29.86	13.51
RTRMN (semantic) [31]	55.26	45.15	39.62	35.87	25.98	55.38	180.25
RTRMN (statistical) [31]	80.28	73.22	68.21	63.93	42.58	77.26	312.70
SAA [30]	79.62	74.01	69.09	64.77	38.59	69.42	294.51
baseline	83.21	76.78	71.09	66.02	42.93	77.63	314.78
structured attention (ours)	**85.38**	**80.35**	**75.72**	**71.49**	**46.32**	**81.41**	**334.89**

* The "-" means that the scores are not reported in the reference papers.

TABLE VI
EVALUATION SCORES (%) OF DIFFERENT METHODS ON THE SYDNEY-CAPTIONS DATA SET [27]

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
VLAD+RNN [28]	56.58	45.14	38.07	32.79	26.72	52.71	93.72
VLAD+LSTM [28]	49.13	34.12	27.60	23.14	19.30	42.01	91.64
mRNN [27]	51.30	37.50	20.40	19.30	18.50	-	161.00
mLSTM [27]	54.60	39.50	22.30	21.20	20.50	-	186.00
mGRU [62]	69.64	60.92	52.39	44.21	31.12	59.17	171.55
mGRU-embedword [30]	68.85	60.03	51.81	44.29	30.36	57.47	168.94
ConvCap [63]	74.72	65.12	57.25	50.12	34.76	66.74	214.84
Soft-attention [28]	73.22	66.74	62.23	58.20	39.42	71.27	**249.93**
Hard-attention [28]	75.91	66.10	58.89	52.58	38.98	71.89	218.19
CSMLF [29]	59.98	45.83	38.69	34.33	24.75	50.18	75.55
SAA [30]	68.82	60.73	52.94	45.39	30.49	58.20	170.52
baseline	73.05	64.37	56.67	52.80	36.50	69.79	215.21
structured attention (ours)	**77.95**	**70.19**	**63.92**	**58.61**	**39.54**	**72.99**	237.91

* The "-" means that the scores are not reported in the reference papers.

TABLE VII
EVALUATION SCORES (%) OF DIFFERENT METHODS ON THE RSICD DATA SET [28]

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
VLAD+RNN [28]	49.38	30.91	22.09	16.77	19.96	42.42	103.92
VLAD+LSTM [28]	50.04	31.95	23.19	17.78	20.46	43.34	118.01
mRNN [27]	45.58	28.25	18.09	12.13	15.69	31.26	19.15
mLSTM [27]	50.57	32.42	23.19	17.46	17.84	35.02	31.61
mGRU [62]	42.56	29.99	22.91	17.98	19.41	37.97	124.82
mGRU-embedword [30]	60.94	46.24	36.80	29.81	26.14	48.20	159.54
ConvCap [63]	63.36	51.03	41.74	34.52	33.25	57.70	166.48
Soft-attention [28]	67.53	53.08	43.33	36.17	32.55	**61.09**	**196.43**
Hard-attention [28]	66.69	51.82	41.64	34.07	32.01	60.84	179.25
CSMLF [29]	51.06	29.11	19.03	13.52	16.93	37.89	33.88
RTRMN (semantic) [31]	62.01	46.23	36.44	29.71	28.29	55.39	151.46
RTRMN (statistical) [31]	61.02	45.14	35.35	28.59	27.51	54.52	148.20
SAA [30]	67.60	54.30	44.33	36.45	31.09	55.36	193.96
baseline	68.39	53.64	43.69	36.37	30.35	55.76	154.80
structured attention (ours)	**70.16**	**56.14**	**46.48**	**39.34**	**32.91**	57.06	170.31

SAA uses the VGG-16 and sound GRUs as its encoder and uses another GRUs as its decoder.

10) Baseline: We first remove the proposed structured attention module of our method and replace it with a standard soft-attention module [21] while keeping other configurations unchanged as the baseline method.

Tables V–VII show the accuracy of our method and the above comparison ones on the three different data sets.

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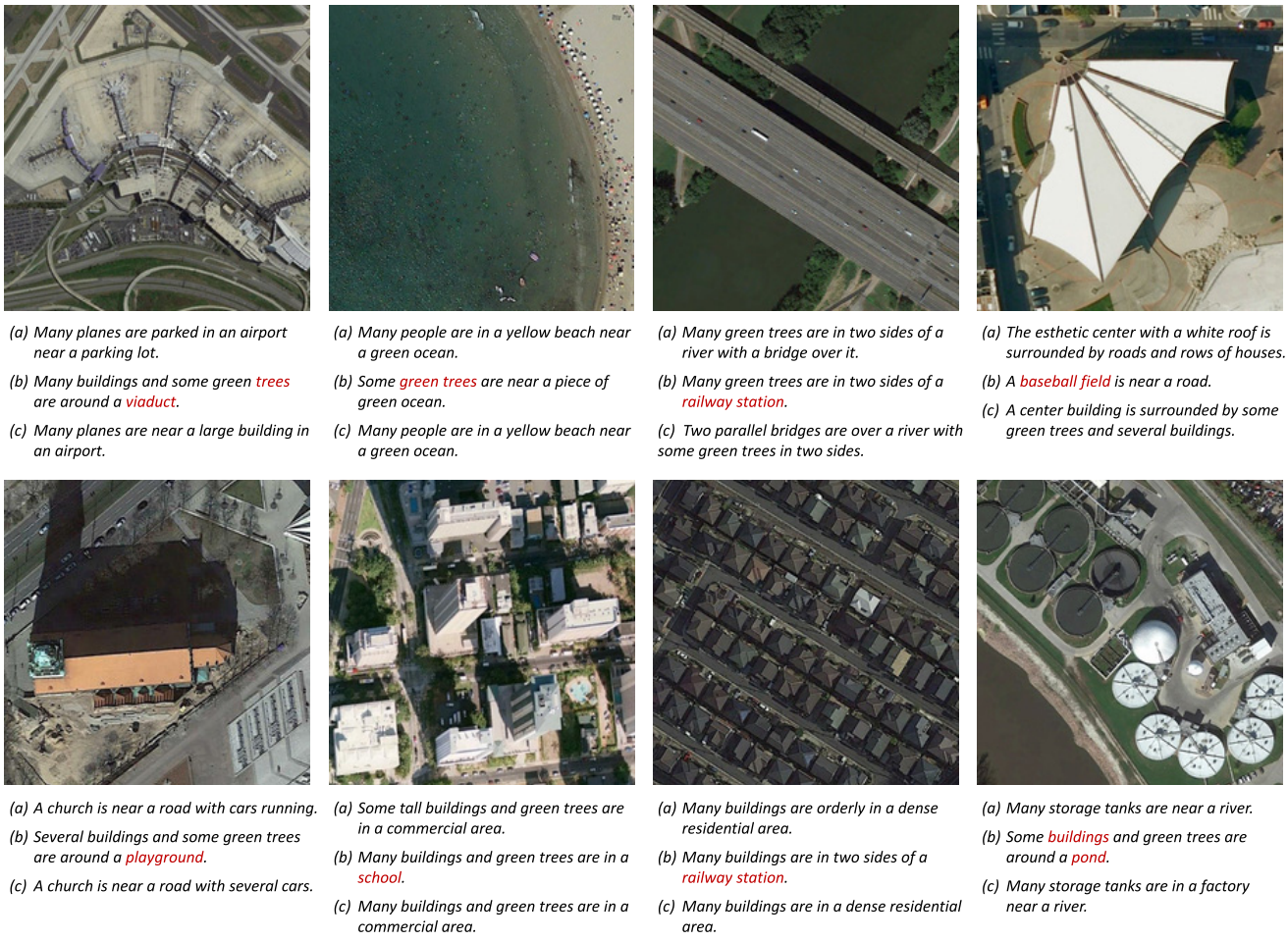


Fig. 7. Captioning results of the baseline method and the proposed method on the RSICD data set [28]. In each group of the image, (a) shows one of the five ground-truth captions, (b) shows the captions generated by the baseline method, and (c) shows the captions generated by the proposed method. The words that do not match the images are marked in red.

All comparison methods follow the same fixed partitions of data (80% for training, 10% for validation, and 10% for test), which makes the comparison fair. In these tables, the best scores are marked as bold. For the comparison methods, the metric scores are taken from the articles that proposed them. Since Qu *et al.* [27] did not report the ROUGE-L scores on UCM-Captions and Sydney-Captions data sets, these numbers are missing in Tables V and VI. We can see our method achieves the best accuracy in most of the entries. For example, on the RSICD data set, our baseline method (Resnet50 + LSTM + soft attention), which also applies beam search with the same beam size during the inference stage, is already better than most of the other methods, as shown in Table VII. When we integrate the structured attention, we further improve our baseline by 2.97% on BLEU-4, 2.56% on METEOR, 1.30% on ROUGE-L, and 15.51% on CIDEr-D. While the baseline and proposed structured attention method both use ResNet-50 as the encoder and LSTM as the decoder, the proposed approach always achieves a score higher than the baseline, which means that the achieved improvement is due to the proposed structured attention method.

E. Qualitative Analysis

1) Caption Generation Results: In Fig. 7, we show some captioning results of our method on the RSICD data set. In each group of the result, we show one of the five ground-truth sentences, the sentence generated by our method (Resnet50 + LSTM + structured attention), and that generated by our baseline (Resnet50 + LSTM + soft attention) accordingly. The generated words that do not match the images are marked in red. As we can see, the baseline method tends to generate false descriptions of small objects or irregularly shaped objects. This is because the regions captured by the soft attention are coarse-grained and unstructured, which leads to insufficient use of structured information and low-level visual information. As a comparison, our method generates more accurate descriptions, which is because the structured attention can fully exploit the structured information of remote sensing semantic contents.

2) Visualization of the Attention Weights: In Fig. 8, we visualize the weights produced by the standard soft attention and the proposed structured attention. For each image, we visualize the attention weights of the attention module when the decoder is generating corresponding words, such as trees,

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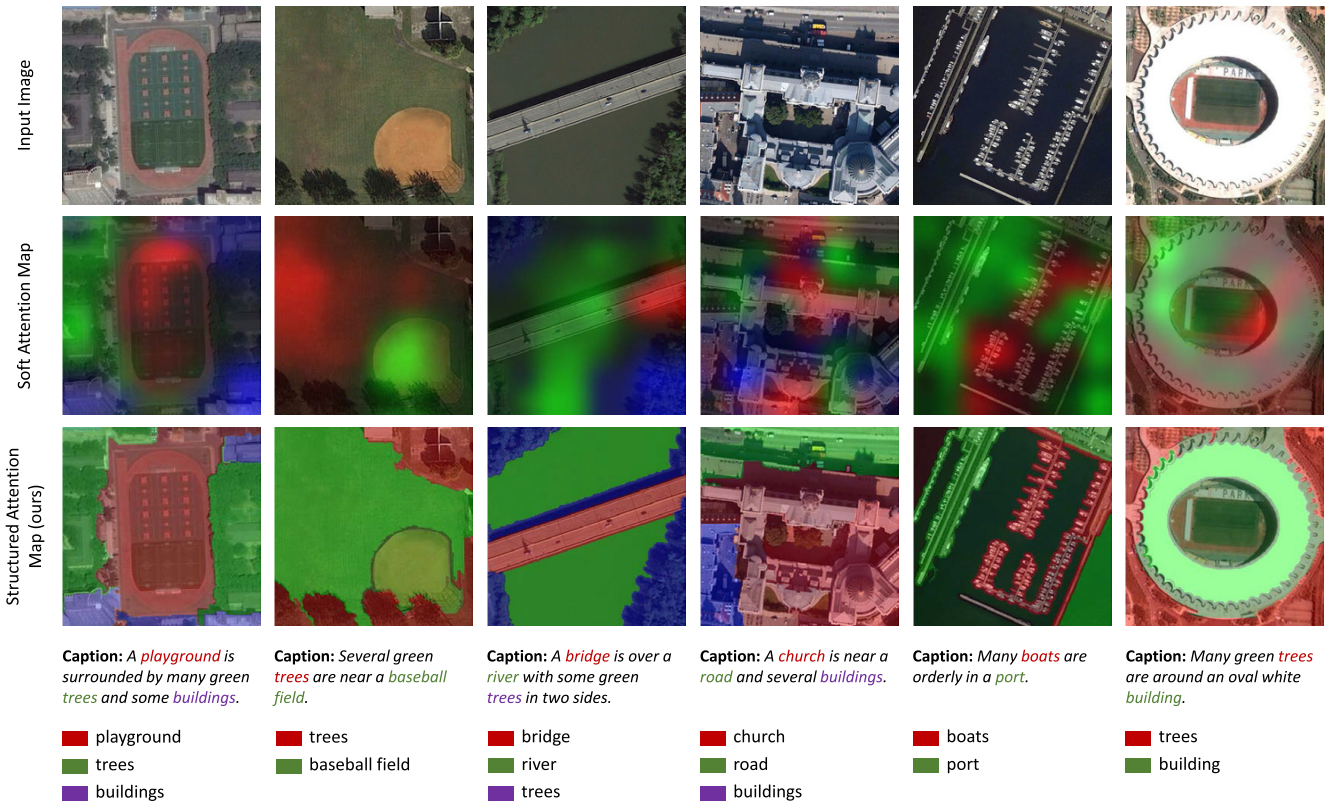


Fig. 8. Visualization of the attention weights on the RSICD data set [28]. The attention weights are displayed in different colors and are overlaid on the original image for a better view. The proposed structured attention-based method produces much more detailed and structural rational attention weights compared to the standard soft attention-based method [21]. To reduce the mosaic effect, the soft attention maps are smoothed by bilinear interpolation multiple times as suggested by previous literature [21].

playgrounds, and buildings. For each word, the attention weights are displayed in different colors and are overlaid on the original image for a better view. As we can see that the structured attention can produce much more detailed and structural rational attention weights compared to standard soft attention. Although we only train our method with image-level annotations and do not use any pixelwise annotations during the training, our method still produces accurate and meaningful segmentation results. The attention maps generated by our method, thus, can be considered as a new way for weakly supervised image segmentation.

3) Visualization of Object Masks: Fig. 9 visualizes the regions in some images where the attention is heavily weighted as the decoder generates specific classes of words, such as the river, bridge, building, and tree. We can also see that some generated object masks usually have specific structures. For example, the river is winding, and the bridge is long and straight. By leveraging the low-level vision from the segmentation proposals with typical structures, the proposed structured attention can help the decoder to generate more accurate captions.

F. Speed Performance

In Table VIII, we report the number of model parameters, the number of floating-point operations (FLOPs), training time, and the inference speed (images per second) of our method. The results are computed on the RSICD data set. The training time and inference speed are tested on an NVIDIA TITAN X (Pascal) graphics card. Comparing with the baseline method, the proposed method does not increase the number of model parameters. This is because the basic network structure does not need to be changed when we modify the standard attention module to the proposed structured attention module. The proposed method even decreases the number of FLOPs because the baseline method generates $14 \times 14 = 196$ groups attention weights, each corresponding to a uniform grid in the image, while the proposed method only need to generate eight groups attention weights since we set the number of structured regions to 8 in our method. We extract and save region proposals of images locally before training the model and directly load them from the local disk during training.

TABLE VIII

COMPARISON BETWEEN OUR METHOD AND THE BASELINE ON THE NUMBER OF PARAMETERS, FLPOS, TRAINING TIME, AND INFERENCE SPEED (IMAGES PER SECOND). ALL RESULTS ARE REPORTED BASED ON THE UCM-CAPTIONS DATA SET

Method	#parameters	FLPOs	training time	inference speed
baseline	34.6M	7.02G	171min	1.16
ours	34.6M	5.58G	175min	1.09

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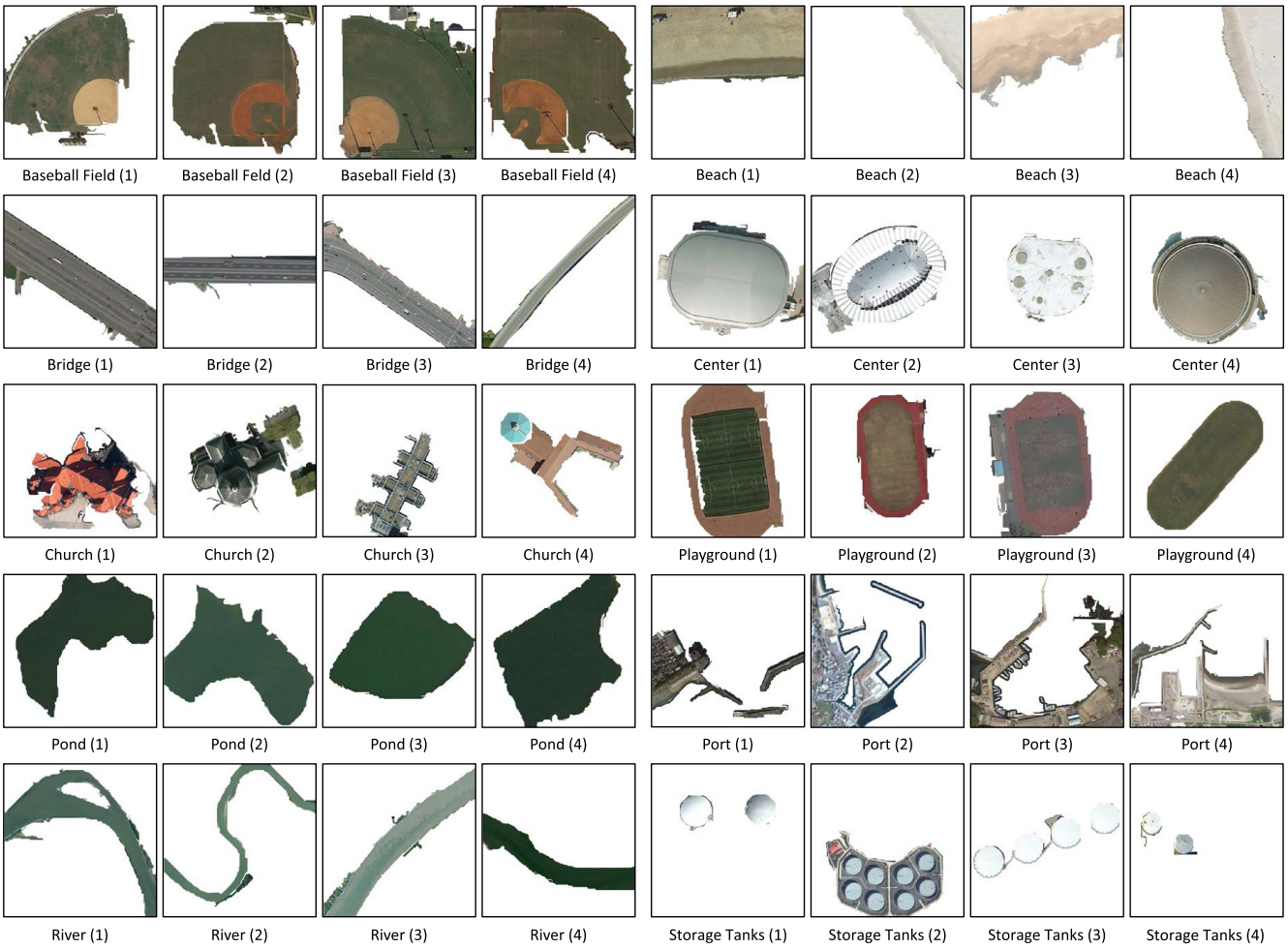


Fig. 9. Visualization of the object masks generated by our method from the RSICD data set. There are ten classes of semantic contents where their masks can be generated by our method, including the baseball field, beach, bridge, center, church, playground, pond, port, river, and storage tanks.

Therefore, the training time is not affected by the region proposal extraction. Since the structured attention module produces only eight structured attention weights instead of 196, which is used in a standard soft attention module, if we do not consider the segmentation time, the inference speed even becomes faster.

IV. Conclusion

We proposed a new image captioning method for remote sensing images based on the structured attention mechanism. The proposed method achieves captioning and weakly supervised segmentation under a unified framework. Different from the previous methods that are based on coarse-grained and soft attentions, we show that the proposed structured attention-based method can exploit the structured information of semantic contents and generate more accurate sentence descriptions. Experiments on three public remote sensing image captioning data sets suggest the effectiveness of our method. Compared with other state-of-the-art captioning methods, our method achieves the best results on most evaluation metrics. The visualization experiments also show that our method can generate much more detailed and meaningful object masks than the soft attention-based method.

Our method also has limitations. Our method is only applicable to high-resolution remote sensing images. If the image is not high resolution, the selective search method will fail to extract available segmentation proposals to support the proposed structured attention mechanism. The number of segmentation proposals used in our structured attention module is fixed. When the input image contains more semantic content than the predefined number of proposals, the structured attention module may fail to focus on the most salient regions. In future work, we will make the number of segmentation proposals adaptive. Another future direction is to combine remote sensing image captioning with object detection. Particularly, we will focus on weakly supervised detection, i.e., to train the detector only based on the sentence annotation.

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